MUHAMMAD ABDUL HADI

Data Science Undergraduate

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PROFILE SUMMARY

Detail-oriented Data Science undergraduate at Pak-Austria Fachhochschule (PAF-IAST) with a strong technical foundation in Python and C++ programming, Object-Oriented Programming (OOP), and data preprocessing. Experienced in engineering end-to-end software simulations and AI-driven monitoring prototypes from scratch. Active campus leader with proven experience in digital media outreach, student residential governance, and community initiatives, seeking international graduate study and technical research opportunities in Data Science.

EDUCATION

Bachelor of Science in Data Science — **2025 – Present (Expected 2029)**

Pak-Austria Fachhochschule: Institute of Applied Sciences and Technology (PAF-IAST)

TECHNICAL SKILLS & PROFICIENCIES

Programming Languages:	Python, C++ (C++11, Object-Oriented Programming, Dynamic Memory Management, File I/O, Templates)
Data & Software Concepts:	Data Collection & Preprocessing, Anomaly Detection Workflows, Polymorphic Design, Exception Handling, Linear Search & Array Structures
Tools & Version Control:	GitHub
Certifications & Productivity:	Completed coursework in Python and C++ Programming; Fast touch-typing speed averaging 85 WPM

TECHNICAL PROJECTS

FloraSentinel AI — Plant Growth Monitoring & Conservation Support — **Python | AI Lab Project**

Role: Data Collection & Preprocessing

- Led data collection and preprocessing workflows combining plant imagery with environmental sensor readings (soil moisture, temperature, humidity, pH, and light intensity) to detect early signs of plant stress.
- Built clean data pipelines to extract visible plant growth trends (leaf area and greenness) and flag anomalous environmental readings against species-specific care rules.
- Evaluated visible-growth estimates against manual measurements and benchmarked anomaly precision, recall, and false-alarm rates against a fixed-threshold baseline for dashboard reporting.

Rural Hospital Emergency Room Simulator — **C++ | OOP Lab Project**

Object-Oriented Architecture & Automated Patient Workflow Simulation

- Engineered a console-based ER simulator in C++ using core OOP principles (inheritance across *Person*, *Patient*, *CriticalPatient*, and *StablePatient* classes, encapsulation, and pure virtual functions) to digitize rural clinic workflows.
- Implemented dynamic memory allocation for scalable waiting-room queue management and runtime polymorphic triage routing to prioritize critical versus stable admissions.
- Developed automated billing and persistent record storage using friend classes (*BillingSystem*), operator overloading (+), generic function templates (`calculateTax<T>` for 5% tax calculation), custom exception handling (*InvalidDataException*), and `<fstream>` file I/O.

Console-Based Movie Management System — **C++**

Data Structures & Search Logic Application

- Developed an interactive command-line application in C++ to store, browse, update, and manage a library of film records.
- Implemented parallel arrays and linear search algorithms to filter and retrieve records accurately with full input validation routines.

LEADERSHIP & EXTRACURRICULAR EXPERIENCE

Deputy Director of Media Affairs — **Data Science Society, PAF-IAST**

- Directed digital outreach and promotional media campaigns for technical workshops, seminars, and student society events.
- Collaborated with society leadership to boost student participation and maintain clear communication across campus channels.

Hostel Committee Member — **PAF-IAST**

- Represented resident students on the hostel committee, addressing daily accommodation, facility maintenance, and student welfare matters.
- Coordinated between hostel administration and residents to resolve operational issues and maintain an organized living environment.

Organizer & Volunteer — **University Community Service Initiatives**

- Organized and participated in a campus-wide environmental cleaning drive and a local animal feeding campaign, coordinating student volunteers and on-ground logistics.